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DESIGN ENGINEER PORTFOLIO

MAYLI JONES

Research

Design

Iterate

Create

2026

About Me

Aspiring Product Design Engineer

I am a Design Engineering student at Imperial College London with a particular interest in user experience, human-centred design and developing intuitive products and services. I enjoy combining user research and creative problem-solving with prototyping and technical development to take ideas from early insights through to tested solutions.

My approach combines technical development with an understanding of how people interact with products. User feedback, ergonomics and design psychology help me create outcomes that are not only functional, but intuitive and well considered.

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Outside Engineering

Outdoors

Canyoning,
Coasteering, Hiking

Sports

Indoor Bouldering
and Swimming

Music

Piano (Grade 8), Violin (Grade 8)
Drums and Singing

Additional

Part-time tutor in Maths, Science and English
Fluent in spoken Mandarin Chinese

Mayli Jones



MindMeter

Measuring the Unmeasurable

The Challenge

Create an interactive mechatronic product using resourceful design, simple electronics and a range of engaging mechanical movements.

The Solution

A playful machine that measures sanity, perception, luck and emotion through sensors, lights and interactive mechanical games, creating a varied user experience.

My Role

- Developed the full CAD model and mechanical systems
- 3D printed, finished and assembled the prototype
- Integrated electronics through soldering and wiring

Key Skills

CAD	3D Printing	Rapid Prototyping
Arduino	DFA	Electronics Integration
Mechanical Design	C++	Soldering



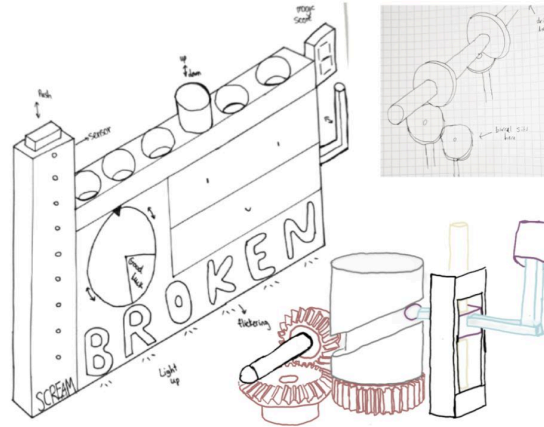
Development Journey

Mechanisms into Motion:

The project developed through rapid prototyping, CAD iteration and repeated trial and error. Rather than following a fixed design from the start, each mechanism was built, tested and refined as we learned how the different moving parts behaved in practice.

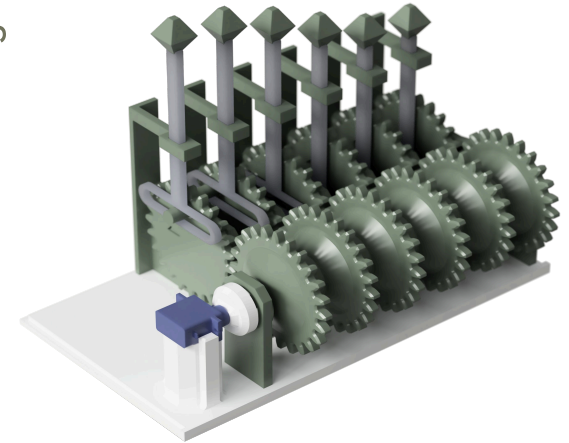
CAD helped us explore gear trains, linkages and internal layouts before manufacturing, while physical prototypes revealed issues with friction, alignment and reliability. These findings guided each new iteration, allowing the mechanical and electronic systems to gradually develop into one integrated interactive machine.

Concept Development



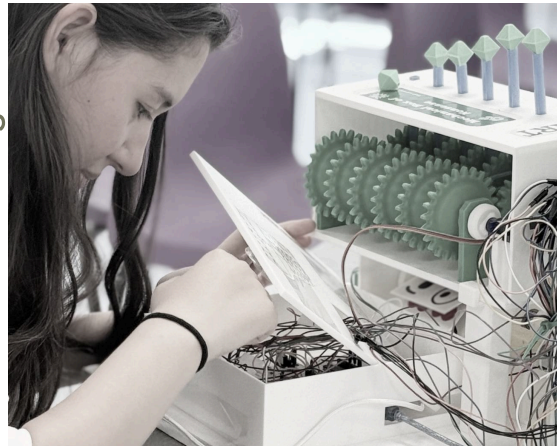
Explored and combined interactive mechanism ideas through sketches.

Mechanical Design



Refined gears, linkages and moving assemblies through CAD and prototypes.

Electronic Integration



Soldered and integrated Arduino, sensors, motors and LEDs.

Public Presentation



Presented and demonstrated at GERF and the 2026 Open Day to visitors.

MindMeter

Sound-Responsive Tower

A push button activates the microphone, which measures sound level and changes the LEDs from green to red.

Capacitive Touch Input

Touching the sensor stops the flickering light.

Hand-Tracking Pistons

A ToF sensor measures hand distance, allowing the crystal-topped pistons to follow the user in a wave-like motion.

Scotch-Yoke Mechanism

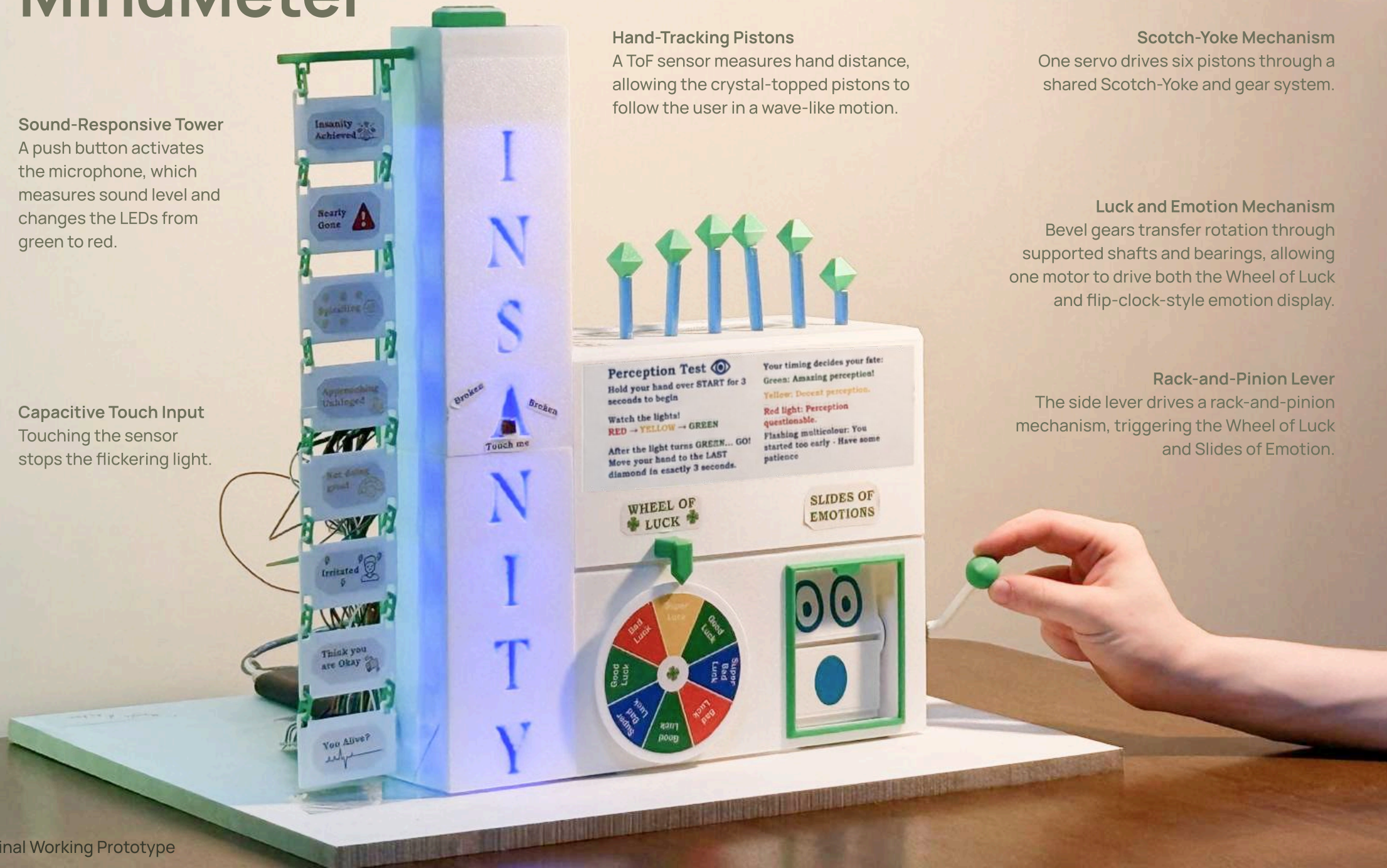
One servo drives six pistons through a shared Scotch-Yoke and gear system.

Luck and Emotion Mechanism

Bevel gears transfer rotation through supported shafts and bearings, allowing one motor to drive both the Wheel of Luck and flip-clock-style emotion display.

Rack-and-Pinion Lever

The side lever drives a rack-and-pinion mechanism, triggering the Wheel of Luck and Slides of Emotion.



SwivelCut

Cardboard Made Even Easier

The Problem

Cardboard prototyping in DT GCSE often depends on craft knives or specialist machinery, making the process unsafe, inconsistent and difficult for students without CAD experience.

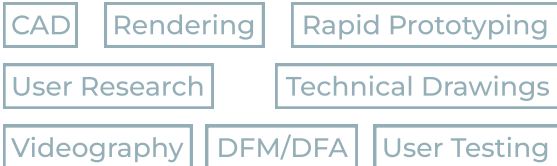
The Solution

SwivelCut follows a student's hand-drawn path to produce cleaner cardboard cuts, making prototyping safer, more accurate and accessible without requiring CAD knowledge.

My Role

- Led CMF, branding and visual direction
- Planned and conducted all user testing and early research
- Produced technical drawings and prototypes

Key Skills



Development Journey

Research to Reality:

The design was developed through user research, engineering analysis and iterative prototyping. Interviews with teachers and students identified key issues around safety, accessibility, accuracy and confidence when cutting cardboard in school workshops.

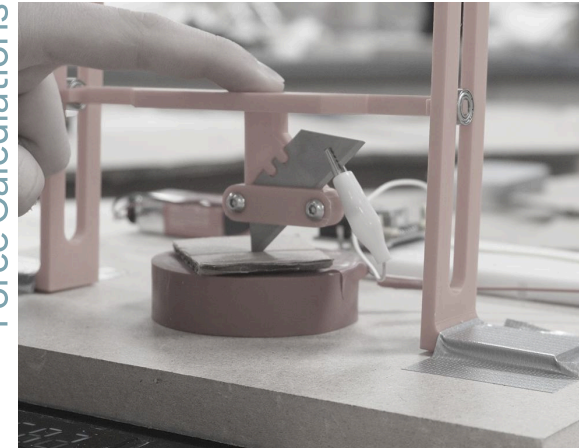
Force testing defined the mechanical requirements, while repeated prototypes refined the arm movement, cutting head, controls and clamping system. Alongside the product, I developed the packaging, user guides, technical drawings and DFMA strategy, while also exploring branding, market positioning, costing and retail viability. Final user testing then validated the setup process and highlighted further opportunities for improvement.

User Research



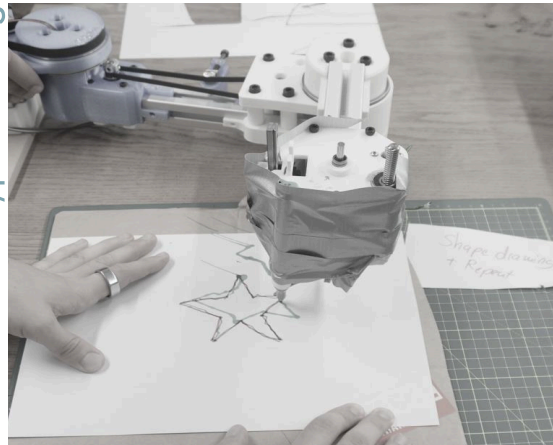
Interviews helped identify issues with safety, accessibility and confidence.

Force Calculations



Piercing and drag tests established the required motor torque and belt ratios.

Prototype Testing



Belt tension, pulley ratios and geometry were refined to improve accuracy.

User Testing



Classroom testing refined setup, clamping, controls and head changes.

SwivelCut

Simplified Controls

Physical buttons and status LEDs provide clear feedback without requiring a screen or CAD interface.

Interchangeable Tool Heads

The tracing and cutting heads attach to the same arm, allowing users to record a path before reproducing it in cardboard.

Stabilisation and Repeat

Smooths the recorded path to better match the user's intent, and can replay it for consistent batch cutting.

Belt-Driven Arm System

Timing belts transfer motor movement through the articulated arms, providing controlled motion across an A3 working area.

Integrated Safety System

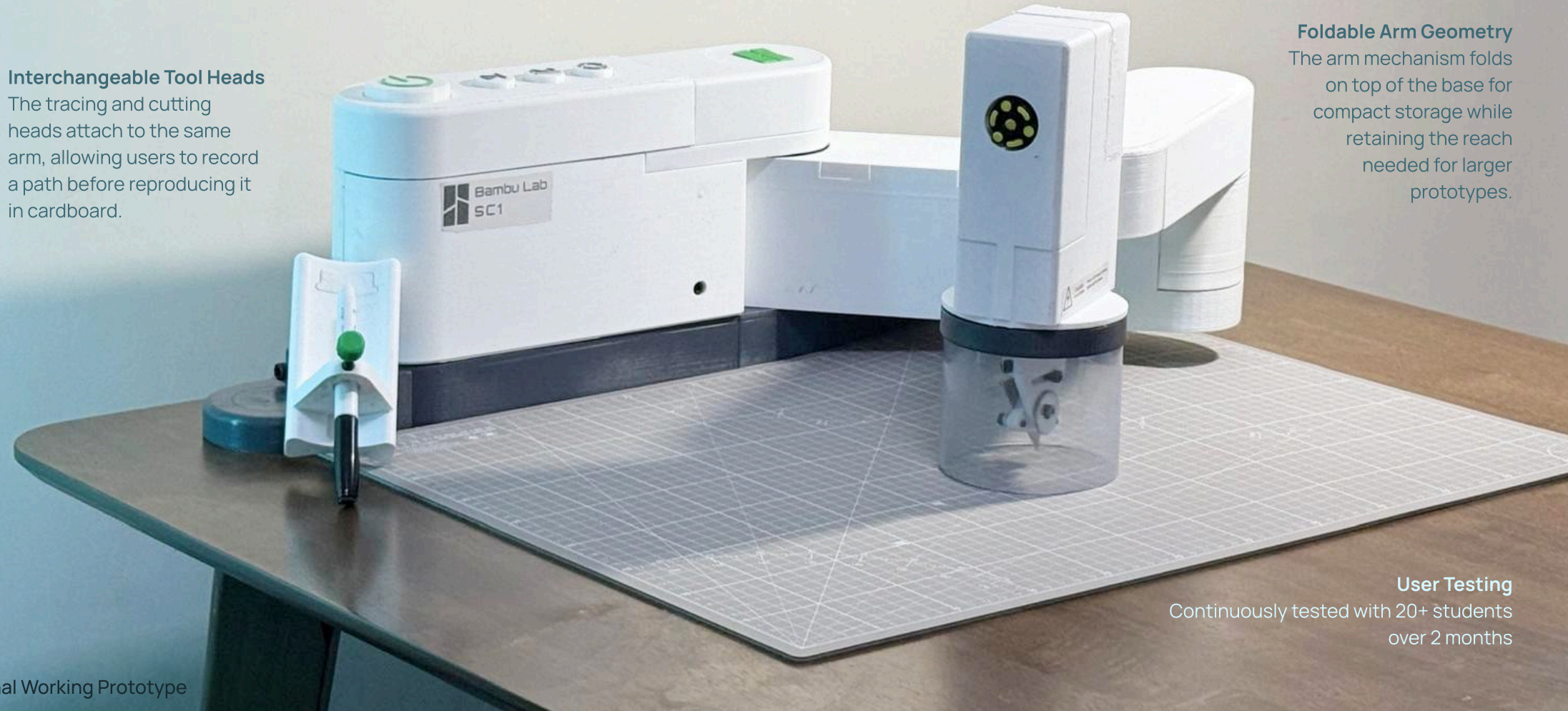
Movement and resistance monitoring stop the machine if unexpected forces or movement are detected during use.

Foldable Arm Geometry

The arm mechanism folds on top of the base for compact storage while retaining the reach needed for larger prototypes.

User Testing

Continuously tested with 20+ students over 2 months



Self Balancing Robot

The Challenge

Starting from a bare robot board built for the module, write the control software to get it balancing upright on two wheels in real time, stable enough to also react live to music.

The Solution

A PID control system keeps the robot upright, while MicroPython beat detection lets it dance without losing balance. I later modelled it in MATLAB to understand its response.

My Role

- Wrote and tuned PID balancing code
- Implemented MicroPython beat detection with a moving average filter
- Modelled the robot as an inverted pendulum in MATLAB

Key Skills

Control Theory

MATLAB

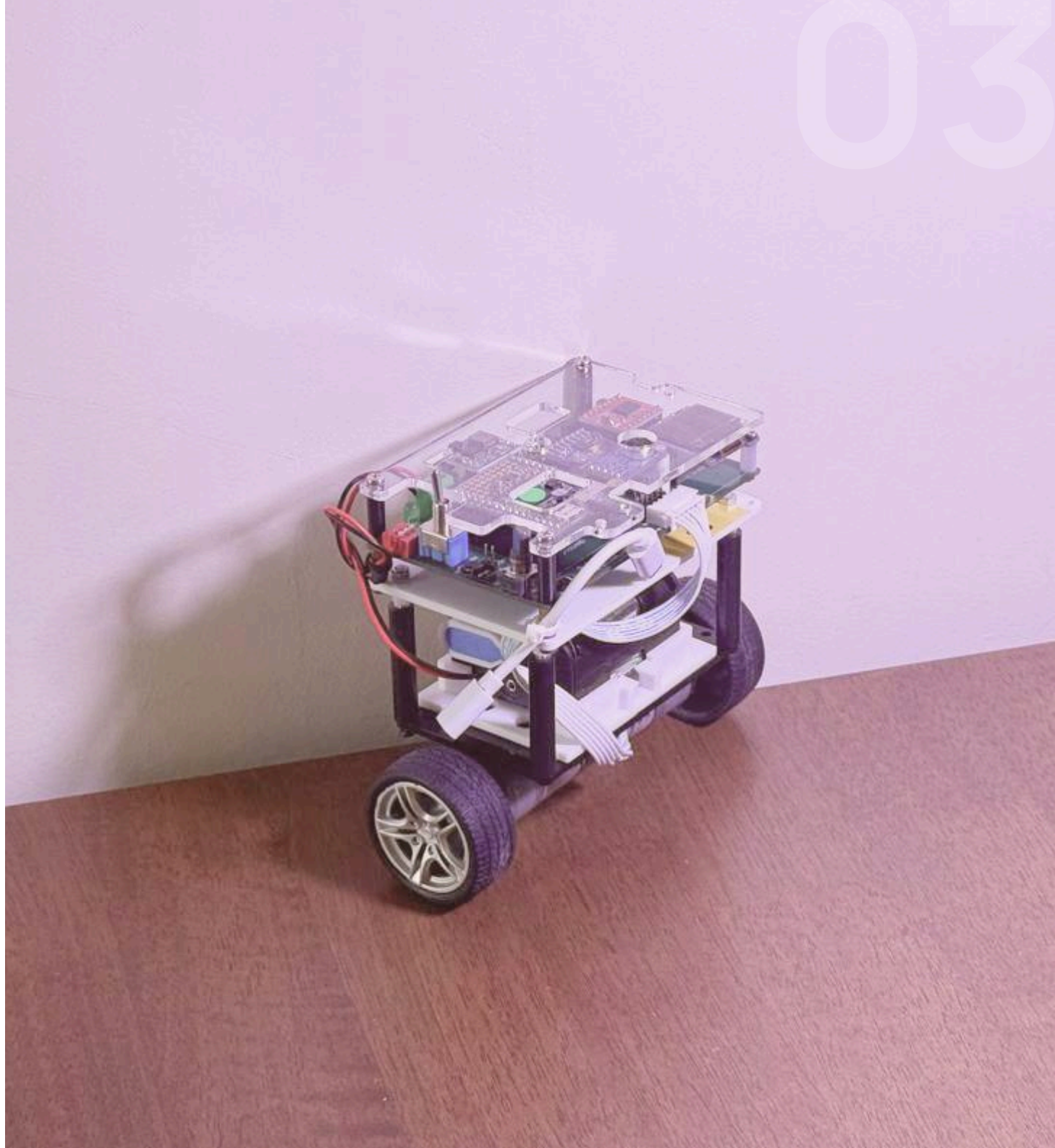
PID Tuning

MicroPython

Signal Processing

Problem Solving

Embedded Systems



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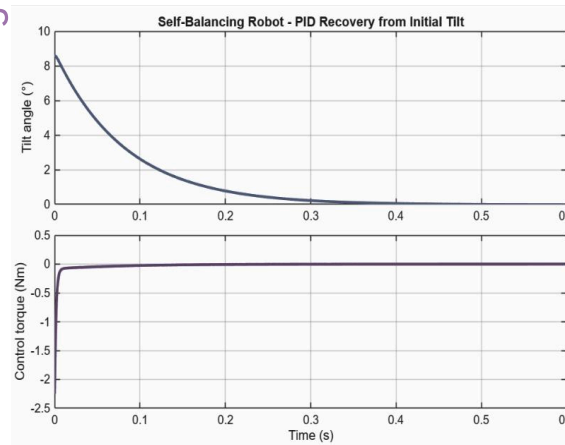
Development Journey

Maths and Music:

Working from a robot board built for the module, we wrote the balancing and beat-detection code separately in MicroPython, tested each on its own, then combined them into one control loop. I tuned the PID gains by hand across different surfaces until the robot recovered smoothly.

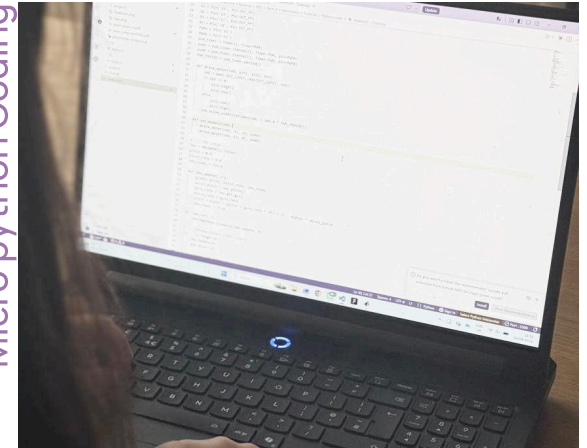
Beat detection used a moving average filter, with a NeoPixel showing beat strength, tested against Billie Jean. Each beat triggers a wheel movement. Afterwards I modelled the robot as an inverted pendulum in MATLAB to understand its response, which recovers from an 8.6° tilt in about 0.4 seconds.

MATLAB Modelling



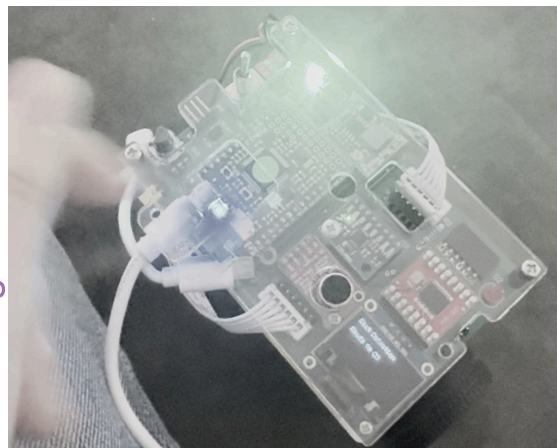
Modelled the robot as an inverted pendulum in MATLAB.

Micro python Coding



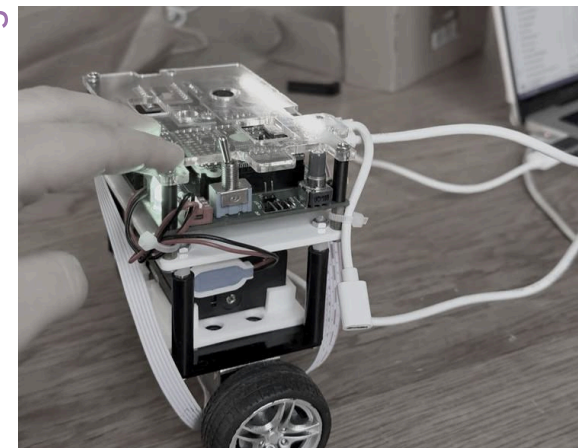
Wrote the balancing and beat-detection logic in MicroPython.

Testing Beat Detection



Refined beat detection using a moving average filter to reliably identify the beat.

PID Tuning



Tuned PID gains across different surfaces, refining stability and response.

Self Balancing Robot

NeoPixel Beat Indicator

Lights up on each detected beat, giving a live visual readout of beat strength during testing.

MicroPython Control Board

Runs the balancing and beat-detection logic directly onboard, no external computer needed.

MATLAB Pendulum Model

Simulated the tilt response after tuning to explain the recovery.

PID Tuning Across Surfaces

Gains refined by testing recovery on different surfaces until the robot could balance smoothly without oscillation.

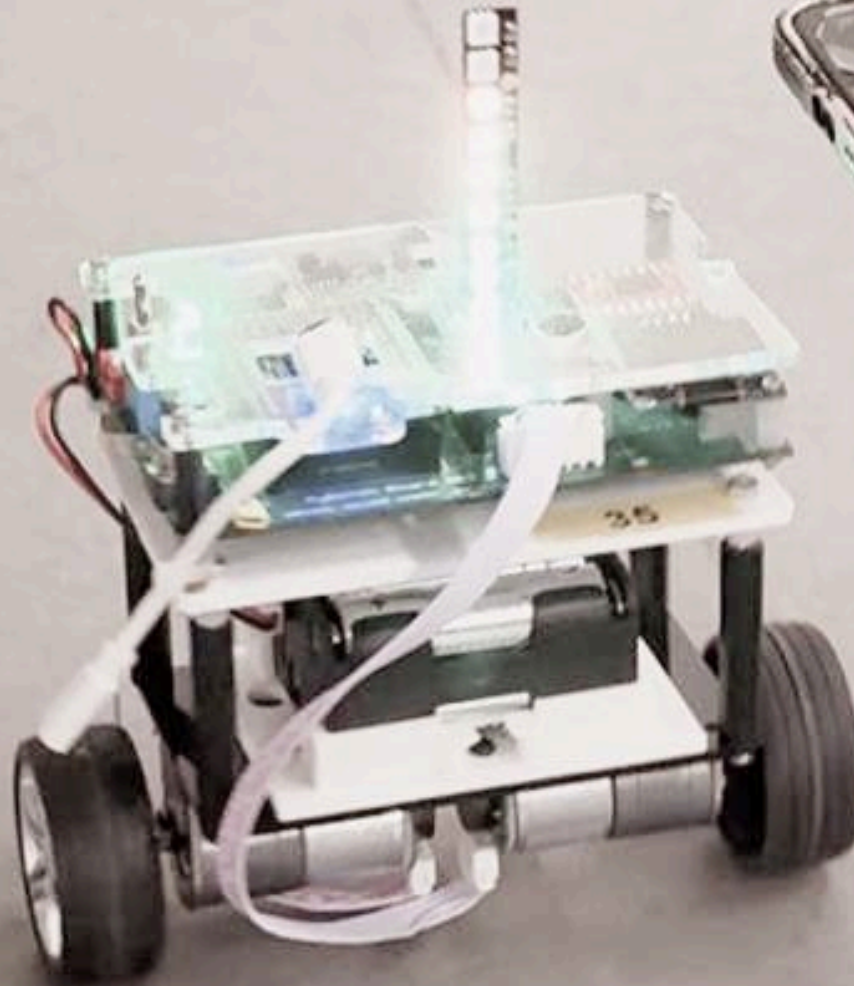
Live Music Input

A phone plays music near the robot, triggering beat-synced movement, left wheel, right wheel, or forwards and backwards.

Moving Average Beat Detection

Filters the incoming audio signal to reliably pick out the beat, tested against tracks like Billie Jean.

Videos of Dancing Segway
[LINK](#)



Modular Family Airplane Seats

The Problem

Economy seats are built around adult anthropometry, leaving children aged 2-4 poorly supported and badly restrained on long-haul flights, driving disruptive behaviour, disturbing other passengers, and increasing stress for parents.

The Solution

Design a modular, size-scalable seat support system that makes long-haul seating safe and comfortable for young children, without requiring a new seat frame or recertification.

My Role

- Applied systems-led design, mapping stakeholder loops and material flows.
- Ideated researched and SWOT-evaluated concept directions
- Calculated environmental impact and assessed operational feasibility

Key Skills

Systems Mapping

Eco Audit

DFMA

User Research

Stakeholder Analysis

Life Cycle Analysis

Sustainable Design



Development Journey

Sustainable Seating:

The project developed through group research before moving into individual concept design. We tore down an existing economy seat, gathering data on materials and environmental sustainability, and carried out a full life cycle analysis, material end-of-life review and eco-audit to understand the seat's environmental footprint.

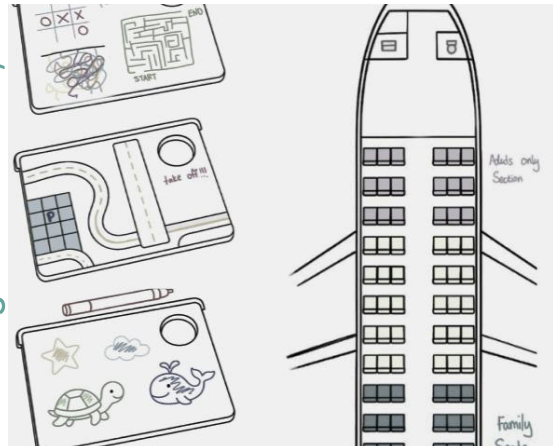
From there, I individually took the concept forward: researching and developing a stakeholder map to understand how different people, from cabin crew to certification bodies to recycling contractors, were affected by the problem, then benchmarking against child-specific design in other industries like Frog Bikes and the Tripp Trapp highchair, before ideating and SWOT-evaluating multiple concept directions and selecting the three-module system. I then redesigned the system map for the new concept and evaluated its social, economic and environmental sustainability against the original.

Economy Seat Teardown



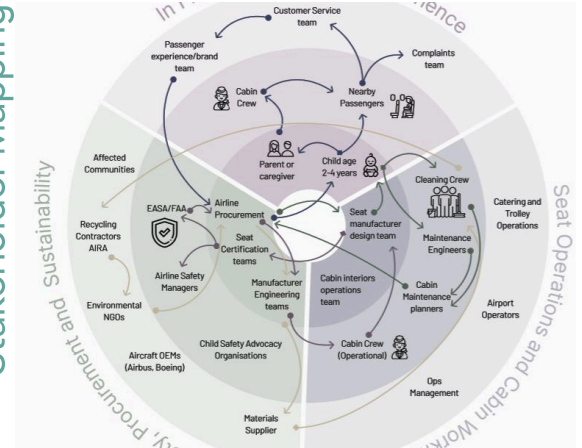
Tore down an existing seat to gather real material and end-of-life data.

Design SWOT Analysis



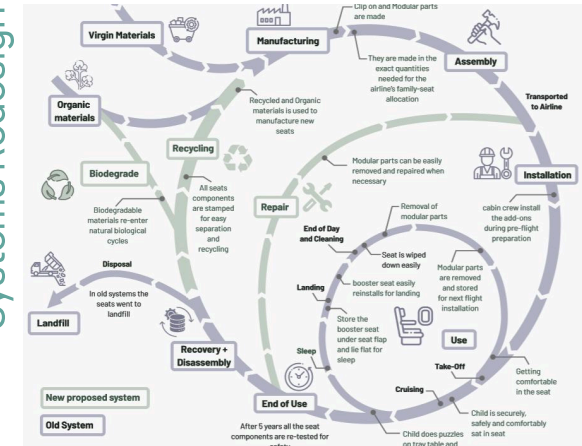
Ideated and SWOT-evaluated directions before selecting the final system.

Stakeholder Mapping



Mapped how the problem affects parents, crew and recycling contractors.

Systems Redesign

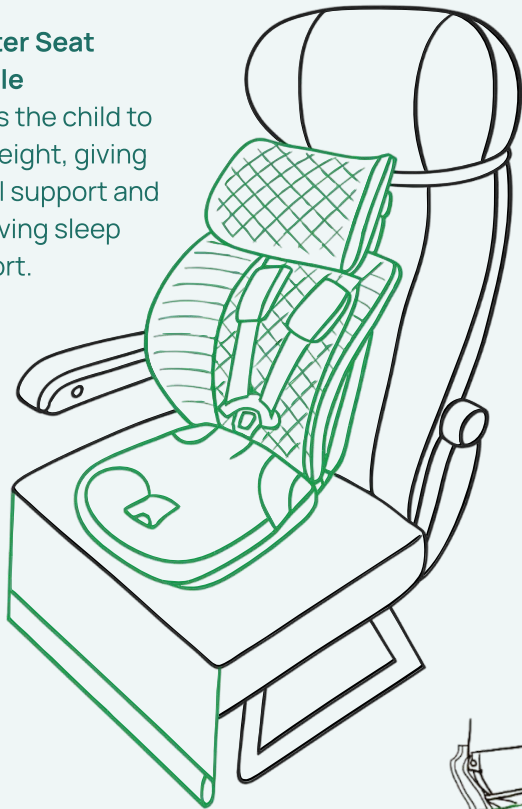


Redesigned the material system to compare circularity against the original.

Modular Family Airplane Seating

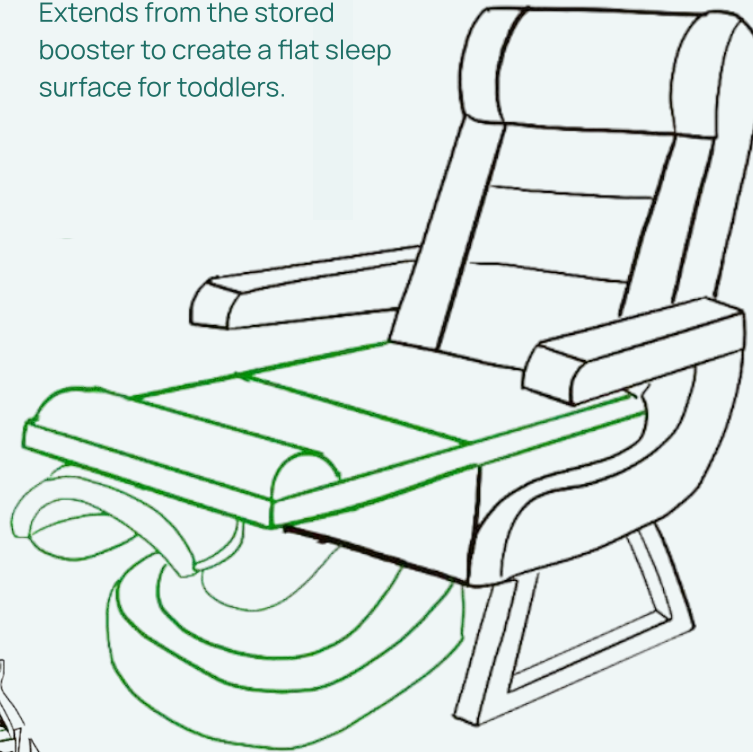
Booster Seat Module

Raises the child to belt height, giving lateral support and improving sleep comfort.



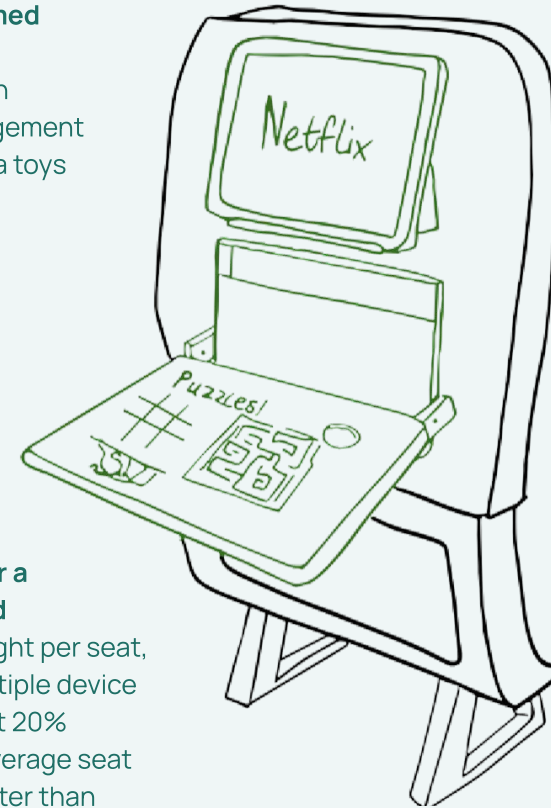
Fold-Out Lie-Flat Surface

Extends from the stored booster to create a flat sleep surface for toddlers.



Puzzle-themed tray table

providing fun tactile engagement without extra toys or pens.



IFE Removed for a Universal Stand

Cuts 4Kg of weight per seat, while fitting multiple device size. Even at just 20% adoption, the average seat stays ~20% lighter than baseline reducing fuel use.

Mesh Material Booster Seat

Mono-material recycled polyester mesh, breathable and simple to recycle.

Anchors Under the Lap Belt

Seatbelt-friendly clip system, with snap fits on the booster seat means no new hardware required.

Under-seat Storage

Folds underneath the seat pan using a table-style locking hinge.

Key risk: module tracking

Storing and tracking boosters between flights is non-trivial, losing or mismatching units could reduce both the economic and environmental benefit.

Roadster Redesign

The Problem

Open-top roadsters can suffer from lift, flow separation and large aerodynamic wakes, reducing high-speed stability. The challenge was to improve downforce without creating a large drag penalty.

The Solution

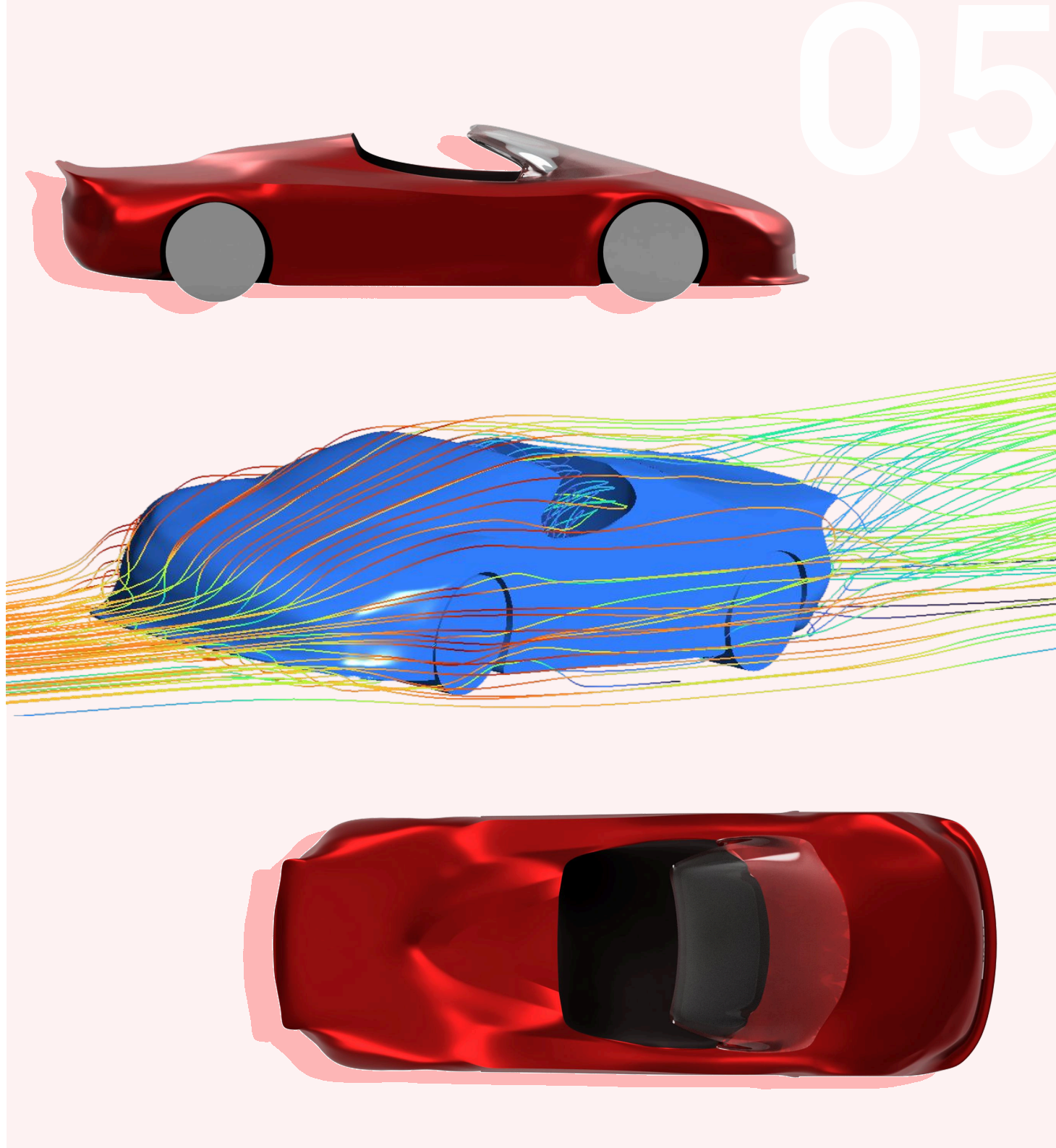
I redesigned the roadster form using aerodynamic research, sketch iteration, CAD and CFD. Simulation results were used to identify flow separation and wake issues, then refine the geometry for improved stability and airflow management.

Individual Project

- Researched and benchmarked roadster aerodynamics and iterated designs
- Built the CAD model and set up CFD simulations
- Analysed results and refined the design from findings

Key Skills

Ansys Fluent	Research	Sketching
CAD	CFD	Problem Solving
Data Analysis	Aerodynamic Analysis	



Development Journey

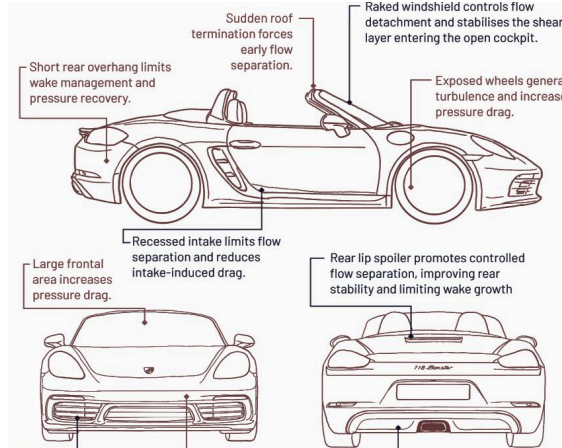
Research and Development

I researched open-top roadster aerodynamics and benchmarked existing vehicles to identify key sources of lift, drag and flow separation, defining the goal of improving stability without a large drag penalty.

I developed concepts through sketching and CAD, then used Ansys Fluent CFD to analyse pressure, velocity, wake formation and airflow.

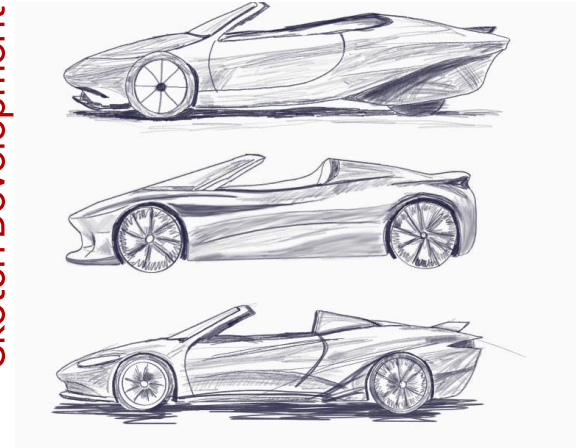
The results highlighted rear flow separation, cockpit recirculation and wake size, which informed further geometry refinements and helped base design decisions on simulation rather than appearance alone.

Technical Research



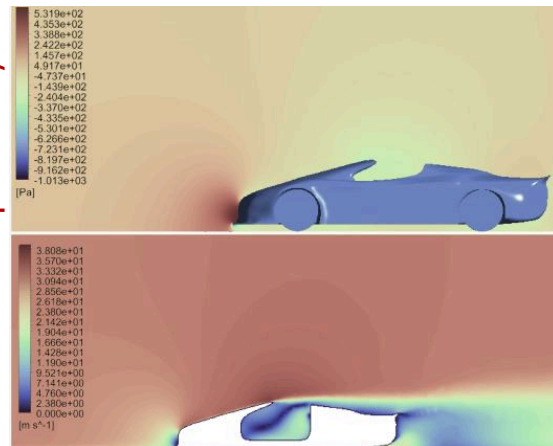
Read research papers and reports about roadsters and aerodynamic design.

Sketch Development



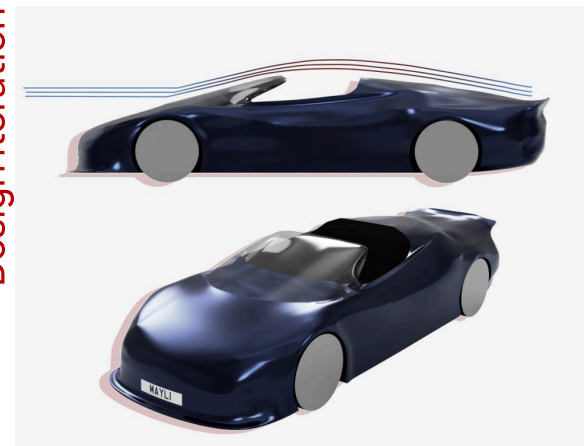
Sketched and Iterated different Ideas for development based off research.

CFD Setup & Analysis



Modelled the car using CAD then performed and analysed the CFD plots

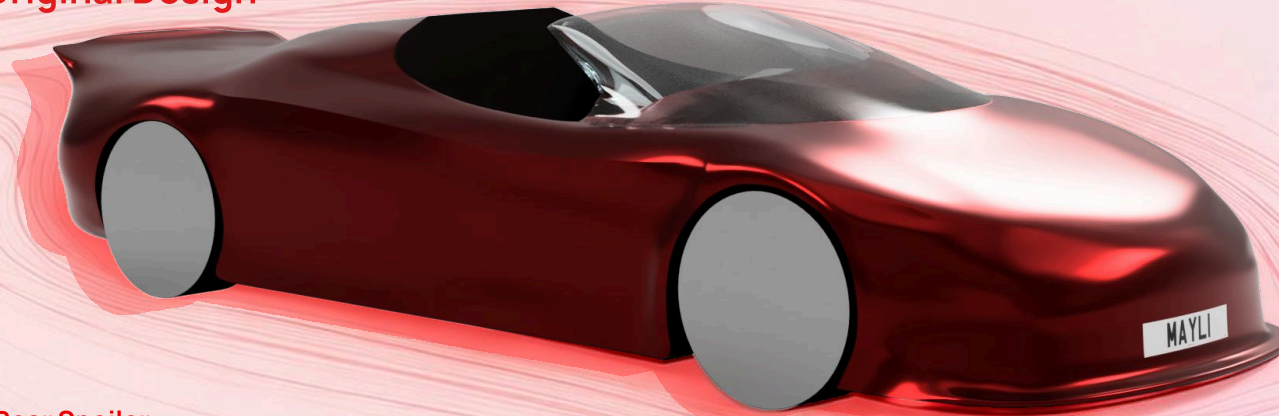
Design Iteration



Made further iterations based of the CFD results to try and improve final results.

Roadster Redesign

Original Design



Cockpit Recirculation

CFD showed a large recirculating vortex inside the open cabin, causing energy loss and disturbed rear flow.

High Front Pressure

A strong stagnation region formed at the front of the car, creating a steep pressure gradient.

Raised Rear Hump

Guides airflow further along the body, delaying separation and reducing rear wake size.

Rear Spoiler

Intended to control rear separation and generate rear downforce.

Splitter Separation

High-speed underbody flow separated beneath the splitter, contributing to turbulence and wake formation.

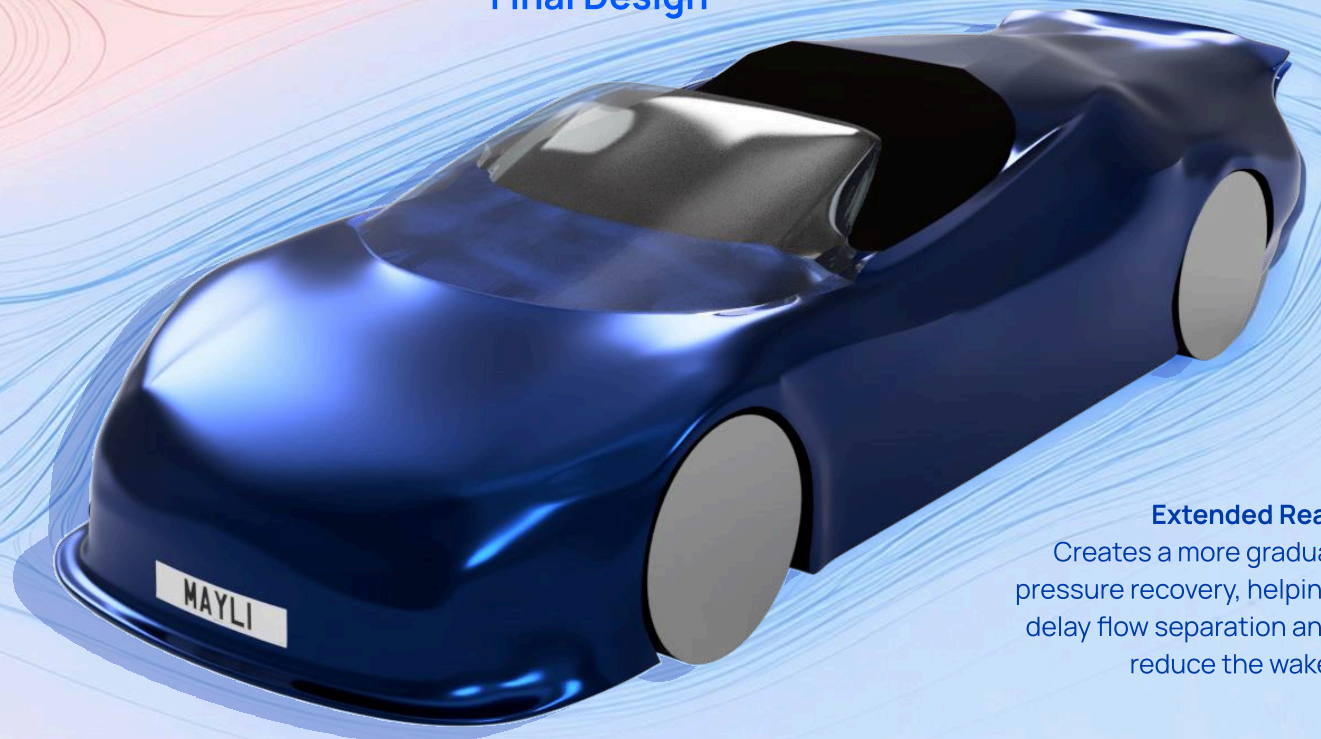
Cabin Edge Lips

Deflect external airflow away from the cockpit, reducing flow entering the cabin.

Weaker Cabin Vortices

Reduced cockpit inflow weakens recirculation, lowering turbulence and creating calmer internal airflow.

Final Design



Extended Rear

Creates a more gradual pressure recovery, helping delay flow separation and reduce the wake.

TiltQuest

Cardboard Made Even Easier

The Problem

People who are new to a neighbourhood can struggle to discover local places, meet others and feel connected to the community.

The Solution

Tilt Quest is a shared interactive table designed for public spaces. Multiplayer games encourage social interaction while introducing users to local shops, food and places to visit.

My Role

- Led project planning and coordinated team tasks throughout development
- Conducted user research and synthesised findings to guide decisions
- Developed low-fidelity prototypes and tested the digital game experience

Key Skills

CAD Rendering Project Management

User Research Concept Development

Team Work Prototyping User Testing



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Development Journey

From People to Prototype

We combined primary and secondary research, including interviews, focus groups, observation, precedent research and affinity mapping, to understand how people build familiarity and connection within a neighbourhood.

The findings informed repeated rounds of concept generation, sketch iteration and team critique, with ideas refined as new insights emerged.

We then developed low-fidelity physical and digital prototypes, tested the game experience, and gathered final feedback from our primary users to identify further improvements and future development opportunities.

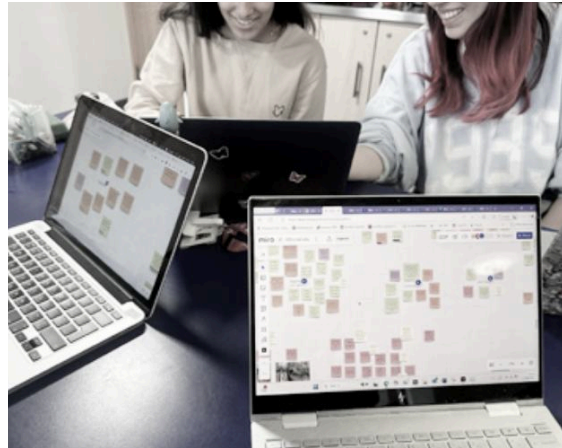
Interactivity

Symbol based - interactive board
could be a map and last refresh the
user's scenarios to help with language barrier

Visual story board
could be a map and last refresh the
user's scenarios to help with language barrier

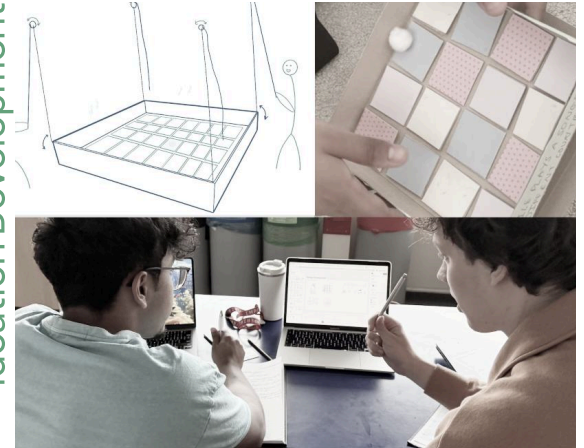
Nonverbal

Initial Research



Conducted Primary and Secondary research and affinity mapped results.

Ideation Development



Sketched and Iterated different ideas then prototyped out of card for testing.

Game Development



Developed and refined the game content and visual system using feedback

Prototype Testing



Tested a low-fidelity prototype with target users to identify improvements.

Tilt Quest

Shared Tilt Platform

Players work together to tilt the board and guide the ball through each challenge.

Two or More Players

The game requires cooperation, encouraging interaction between people who may not know each other.

Local Discovery Content

Games introduce nearby shops, food, landmarks and places to visit around the neighbourhood.

Public-Space Installation

Designed as a shared community experience that can sit within neighbourhood spaces and encourage spontaneous participation.



Pulley-Controlled Movement

Each player controls one side of the suspended board, making communication and teamwork part of the gameplay.